4.4 Factoring Polynomials

Name _____ Period ____ Date _____

Algebra II 4.4 (30-minute plan) · Student sheet

I can…

- factor polynomials by factoring out a monomial and by grouping.
- factor sums and differences of cubes and polynomials in quadratic form.
- use the Factor Theorem.

Part 1 · Notes: fill in the blanks

$a^3 + b^3 =$ _____

$a^3 - b^3 =$ _____

Factor Theorem: $x - k$ is a factor exactly when _____.

Part 2 · Examples (work them with the class)

Example 1

Factor completely.

a. $x^3 + 27$ b. $8x^3 - 1$

Example 2

Factor completely: $x^3 - 3x^2 - 4x + 12$

Part 3 · Partner practice

Partner practice

Factor $x^3 - 64$.